

INFLUENCES OF ANISOTROPY ON BAND GAPS OF 2D PHONONIC CRYSTAL^{★★}

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ABSTRACT Band gaps of 2D phononic crystal with orthotropic cylindrical fillers embedded in the isotropic host are studied in this paper. Two kinds of periodic structures, namely, the square lattice and the triangle lattice, are considered. For anisotropic phononic crystal, band gaps not only depend on the periodic lattice but also the angle between the symmetry axis of orthotropic material and that of the periodic structure. Rotating these cylindrical fillers makes the angle changing continuously; as a result, pass bands and forbidden bands of the phononic crystal are changed. The plane wave expansion method is used to reduce the band gap problem to an eigenvalue problem. The numerical example is given for YBCO/Epoxy composites. The location and the width of band gaps are estimated for different rotating angles. The influence of anisotropy on band gaps is discussed based on numerical results.

KEY WORDS phononic crystal, orthotropy, band gap, periodicity, plane wave expansion

I. INTRODUCTION

Phononic crystals, also called acoustic band gap materials, are composite elastic medium composed of periodic repetitions of two different solids. The acoustic band gaps between the dispersive curves result from large contrast in the elastic constants and mass densities of component materials. The potential application of the band gap property of the phononic crystal attracts wide attention, so many computation methods have been developed^[1–7], for example, the transfer matrix method^[1], the plane wave expansion method^[2,3], the wavelet-based method^[4], the multiple scattering method^[5,6], the finite time domain difference method^[7] and the finite element method^[8]. Each of them has its advantages and disadvantages. The finite element method and the finite difference method have the best applicability. They can be used to compute not only the band structure but also the reflection and the transmission in the phononic crystal with a boundary. However, it is not convenient to draw physical insight as purely numerical methods. The multi-scattering method can provide the best physical insight into the scattering problem. But the formulation is usually very complicated and forbidding. The transfer matrix method is usually used in one dimensional phononic crystal. The multi-scattering method is applied to two dimensional and three dimensional phononic crystals. The plane wave expansion method and the wavelet-based method can be applied to one, two and three dimensional phononic crystals. In addition, the multi-scattering method has the highest computational efficiency. The plane wave expansion method has convergent problems for the large contrast of material constants and for solid/liquid system. When the multi-scattering method is not valid for anisotropic phononic crystals, the plane wave expansion

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method and the wavelet based method are still valid. By using the above-mentioned computational methods, the influence on band gaps of the topology structure, the filling fraction, the lattice array and the mismatch of material constants were also extensively investigated. It is found that the band gap distribution is mainly governed by the lattice array and component materials. However, how to design phononic crystals with tunable band gaps is still an interesting and challenging problem. Goufax (2001) studied the pass band and forbidden band of 2D phononic crystal composed of rigid square columns embedded in air^[9]. The dependence of band gaps on rotating angle was investigated; based on this, a new idea to design phononic crystal is given. Wu and Liu (2002) studied another tunable phononic crystal, namely, square column of water embedded in mercury^[10]. Band gaps were estimated for different rotating angle and filling fraction. Numerical results show that the influence of rotating is more evident when the filling fraction increases. In their works, changing of the band gap structure was carried out by changing the geometry configuration. However, band gaps can also be tuned without changing the geometry configuration. Hou et al. (2003) replaced isotropic square columns with anisotropic cylinders and studied the 2D phononic crystal composed of parallel Pb (Lead) cylinders embedded in PVC (Poly-vinyl-chloride) host^[11]. When rotating these parallel cylinders, the band gap structure is changed accordingly due to the anisotropy of cylinders. The influence of rotating angle and filling fraction were investigated and discussed based on numerical results. In order to show the influence of anisotropy better, in this paper, the Pb cylinders of cubic crystal is replaced by the orthotropic YBCO cylinders. This is because the anisotropic property of material can be embodied more typically by the orthotropic material rather than the cubic crystal material. In addition, two lattice arrays, namely, the square lattice and the triangle lattice, are investigated. The dependence of band gaps upon the rotating angle is obtained numerically. Some new phenomena are observed and explanations are given.

II. DISPERSION EQUATIONS OF 2D PHONONIC CRYSTAL

According to the elastic theory of continuum, the motion equation can be written as

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \mathbf{T} \quad (1)$$

or in the component form

$$\rho \frac{\partial^2 u_i}{\partial t^2} = T_{ij,j} \quad (2)$$

where ρ is the mass density, $\mathbf{u}$ the displacement vector, $\mathbf{T}$ the stress tensor.

Inserting the general Hooke's law

$$T_{ij} = c_{ijkl} u_{k,l} \quad (3)$$

into Eq.(2) leads to

$$\rho \frac{\partial^2 u_i}{\partial t^2} = (c_{ijkl} u_{k,l}),_j \quad (4)$$

where c_{ijkl} is elastic constant of the material. In the case of harmonic motion, the displacement field can be expressed as $u_i = u_i(\mathbf{r})e^{i\omega t}$. However, in a periodic structure, the displacement field can be further expressed as

$$u_i(\mathbf{r}, t) = u_i^k(\mathbf{r})e^{i(kr+\omega t)} \quad (5)$$

where $\mathbf{k}$ is the wave number. $u_i^k(\mathbf{r})$ is a periodic function, and therefore can be expanded into Fourier series. Then, the displacement field is finally expressed as

$$u_i(\mathbf{r}, t) = e^{i\omega t} \sum_G u_i^{k+G} e^{i(k+G)r} \quad (6)$$

In Eq.(6), u_i^{k+G} is Fourier expansion coefficients and $\mathbf{G}$ is the reciprocal lattice vector. Similarly, the mass density ρ and the elastic constant c_{ijkl} can be expressed in Fourier series due to the periodicity of their spatial distribution

$$\gamma(\mathbf{r}) = \sum_G \gamma^G e^{iGr} \quad (7)$$

where γ can be the mass density ρ or the elastic constant c_{ijkl} . γ^G is the Fourier expansion coefficient and can be obtained by

$$\gamma^G = \begin{cases} \gamma^e f + \gamma^m (1 - f) & (\mathbf{G} = 0) \\ (\gamma^e - \gamma^m) P(\mathbf{G}) & (\mathbf{G} \neq 0) \end{cases}$$

In the above equation, the superscript m and e denotes the host material and the filler, respectively. f is the filling fraction. $P(\mathbf{G})$ is called the structure constant which is only dependent of the shape of the filler. In the case of cylindrical filler,

$$P(\mathbf{G}) = \frac{2f \cdot J_1(\mathbf{G}\mathbf{r})}{\mathbf{G}\mathbf{r}} \quad (8)$$

where $\mathbf{r}$ is the radius of cylindrical filler and $J(\cdot)$ the Bessel function.

Inserting Eqs.(6) and (7) into Eq.(4) leads to

$$\omega^2 \sum_{\mathbf{G}'} \rho_{\mathbf{G}-\mathbf{G}'} u_i^{k+\mathbf{G}'} = \sum_{\mathbf{G}'} c_{ijkl}^{G-G'} (\mathbf{k} + \mathbf{G}')_l (\mathbf{k} + \mathbf{G})_j u_k^{k+\mathbf{G}'} \quad (9)$$

Equation (9) is the motion equation in frequency domain. Considering the symmetry of elastic constants

$$c_{ijkl} = c_{ijlk}, \quad c_{ijkl} = c_{jikl} \quad (10)$$

The elastic constant tensor of fourth order, c_{ijkl} , can be reduced to second order, c_{mn} , by index reduction as following:

$$xx \rightarrow 1, \quad yy \rightarrow 2, \quad zz \rightarrow 3, \quad yz/zy \rightarrow 4, \quad zx/xz \rightarrow 5, \quad xy/yx \rightarrow 6$$

Thus, the elastic tensor can be expressed by a matrix with 6×6 dimensions. For isotropic material, only two constants are independent. For cubic crystal material, there are three independent elastic constants. However, there are nine independent elastic constants for orthotropic material. Therefore, the orthotropic material can show the anisotropic property better than the cubic crystal material.

The reciprocal lattice vector $\mathbf{G}$ and $\mathbf{G}'$ ($\mathbf{G} = n_1 \mathbf{b}_1 + n_2 \mathbf{b}_2$, $\mathbf{b}_1$ and $\mathbf{b}_2$ are reciprocal lattice basic vectors) in Eq.(9) should be all possible value in the reciprocal lattice vector space. But, in actual computation, only finite reciprocal lattice vectors are involved, namely, $-N \leq n_1, n_2 \leq N$ (N is a finite integer). For waves propagating in oxy plane, Eq.(9) can be rewritten in the matrix form

$$\omega^2 \begin{bmatrix} \mathbf{R} & & \\ & \mathbf{R} & \\ & & \mathbf{R} \end{bmatrix} \begin{Bmatrix} \mathbf{UX} \\ \mathbf{UY} \\ \mathbf{UZ} \end{Bmatrix} = \begin{bmatrix} \mathbf{XM}_1 & \mathbf{YM}_1 & \mathbf{ZM}_1 \\ \mathbf{XM}_2 & \mathbf{YM}_2 & \mathbf{ZM}_2 \\ \mathbf{XM}_3 & \mathbf{YM}_3 & \mathbf{ZM}_3 \end{bmatrix} \begin{Bmatrix} \mathbf{UX} \\ \mathbf{UY} \\ \mathbf{UZ} \end{Bmatrix} \quad (11)$$

where the explicit expression of $\mathbf{R}$, $\mathbf{XM}_i$, $\mathbf{YM}_i$, $\mathbf{ZM}_i$, $\mathbf{UX}$, $\mathbf{UY}$, $\mathbf{UZ}$ ($i = 1, 2, 3$) are listed in the appendix. After simple transformation, Eq.(11) becomes an eigenvalue equation of $3(2N + 1)$ order

$$\omega^2 \mathbf{X} = \mathbf{A} \mathbf{X} \quad (12)$$

where

$$\mathbf{X} = \{\mathbf{UX}, \mathbf{UY}, \mathbf{UZ}\}^T$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{R}^{-1} \mathbf{XM}_1 & \mathbf{R}^{-1} \mathbf{YM}_1 & \mathbf{R}^{-1} \mathbf{ZM}_1 \\ \mathbf{R}^{-1} \mathbf{XM}_2 & \mathbf{R}^{-1} \mathbf{YM}_2 & \mathbf{R}^{-1} \mathbf{ZM}_2 \\ \mathbf{R}^{-1} \mathbf{XM}_3 & \mathbf{R}^{-1} \mathbf{YM}_3 & \mathbf{R}^{-1} \mathbf{ZM}_3 \end{bmatrix}$$

The relation between the eigenfrequency ω and the wave number $\mathbf{k}$, namely, the dispersive curve, can be obtained by solving the eigenvalue equation. The frequency band where dispersive curve exists is called pass band whereas the frequency band where the dispersive curve does not exist is called forbidden band or band gap.

In the case that orthotropic fillers are embedded in isotropic host medium, the submatrix $\mathbf{ZM}_1$, $\mathbf{ZM}_2$, $\mathbf{XM}_3$ and $\mathbf{YM}_3$ in Eq.(11) are zero. Then, Eq.(12) is separated into two equations.

$$\omega^2 \mathbf{X}_1 = \mathbf{A}_1 \mathbf{X}_1 \quad (13)$$

$$\omega^2 \mathbf{UZ} = \mathbf{A}_2 \mathbf{UZ} \quad (14)$$

where $\mathbf{X}_1 = \{\mathbf{UX}, \mathbf{UY}\}$, $\mathbf{A}_1 = \begin{bmatrix} \mathbf{R}^{-1}\mathbf{XM}_1 & \mathbf{R}^{-1}\mathbf{YM}_1 \\ \mathbf{R}^{-1}\mathbf{XM}_2 & \mathbf{R}^{-1}\mathbf{YM}_2 \end{bmatrix}$, $\mathbf{A}_2 = \mathbf{R}^{-1}\mathbf{ZM}_3$. Eigen-modes obtained from Eqs.(13) and (14) are xy modes and z modes, respectively. In the case of cubic crystal, xy modes and z modes are decoupled too. However, in triclinic and monoclinic crystal, the xy modes and z modes are coupled completely.

III. COORDINATE TRANSFORMATION OF ORTHOTROPIC ELASTIC CONSTANTS

Although the elastic constant tensor is not dependent upon the reference frame, components of the elastic constant tensor are dependent upon the specific coordinate system. Elastic constants of anisotropic material are usually given with main axes of material as reference frame (hereafter, it is called the material coordinate system (x', y', z')). The phononic crystal, as a composite material with periodic structure, needs a coordinate system to show the lattice array. This coordinate system is called the lattice coordinate system (x, y, z) . When rotating the cylindrical filler, the material coordinate system is not superposed with the lattice coordinate system anymore. Elastic constant components in the material coordinate system is denoted by c'_{mn} and that in the lattice coordinate system by c_{mn} . c_{mn} can be obtained from c'_{mn} by coordinate transformation.

$$\mathbf{c} = \mathbf{M}\mathbf{c}'\mathbf{M}^t \quad (15)$$

If the rotating angle is θ (with z axis as rotating axis), then

$$\mathbf{M} = \begin{bmatrix} \cos^2 \theta & \sin^2 \theta & 0 & 0 & 0 & \sin 2\theta \\ \sin^2 \theta & \cos^2 \theta & 0 & 0 & 0 & -\sin 2\theta \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos \theta & -\sin \theta & 0 \\ 0 & 0 & 0 & \sin \theta & \cos \theta & 0 \\ -\sin \theta \cos \theta & \sin \theta \cos \theta & 0 & 0 & 0 & \cos 2\theta \end{bmatrix} \quad (16)$$

After rotating, the elastic constant tensor of orthotropic filler has the form

$$\mathbf{c} = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & c_{16} \\ c_{12} & c_{22} & c_{23} & 0 & 0 & c_{26} \\ c_{13} & c_{23} & c_{33} & 0 & 0 & c_{36} \\ 0 & 0 & 0 & c_{44} & c_{45} & 0 \\ 0 & 0 & 0 & c_{45} & c_{55} & 0 \\ c_{16} & c_{26} & c_{36} & 0 & 0 & c_{66} \end{bmatrix}$$

where c_{44} , c_{55} and c_{45} are involved in the motion equation of z mode. Because c_{44} , c_{55} and c_{45} are dependent upon the rotation angle θ , the rotating will cause the change of wave bands of z mode in the case of orthotropic fillers. However, in the case of cubic crystal fillers, $c_{45} = 0$ and both c_{44} and c_{55} are independent of θ . Therefore, rotating fillers does not influence z mode.

IV. NUMERICAL RESULTS AND DISCUSSIONS

Consider a phononic crystal composed of cylindrical fillers (YBCO) embedded in the host material (Epoxy) in square lattice array and triangle lattice array, see Fig.1. Elastic constants and mass densities of the filler and the host material are listed in Table 1.

Due to the symmetry of the crystal, only those wave vectors $\mathbf{k}$ in the first Brillouin zone, further, in the boundary of first Brillouin zone, are considered in the band structure computation. The first Brillouin zones of square lattice array and triangle lattice array are shown in Fig.2. The wave band

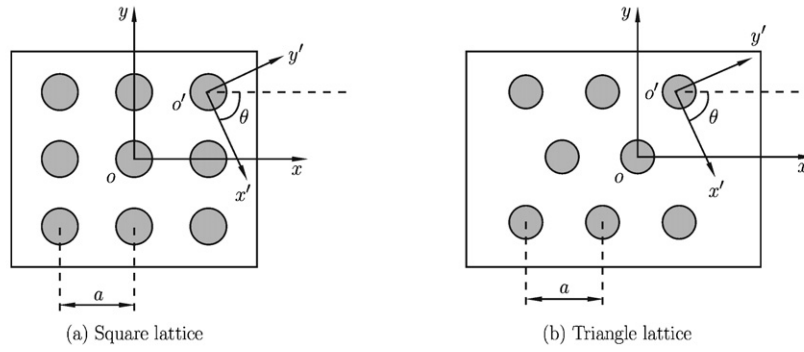


Fig. 1. The cross section of 2D phononic crystal and the rotating angle of cylindrical fillers.

Table 1. Elastic constants and mass densities of components

Component material	Elastic constants (GPa)									Density ρ (kg/m ³)
	c_{11}	c_{12}	c_{13}	c_{22}	c_{23}	c_{33}	c_{44}	c_{55}	c_{66}	
YBCO	268	132	95	231	71	186	37	49	95	6333
Epoxy	7.54	4.57	4.57	7.54	4.57	7.54	1.48	1.48	1.48	1142

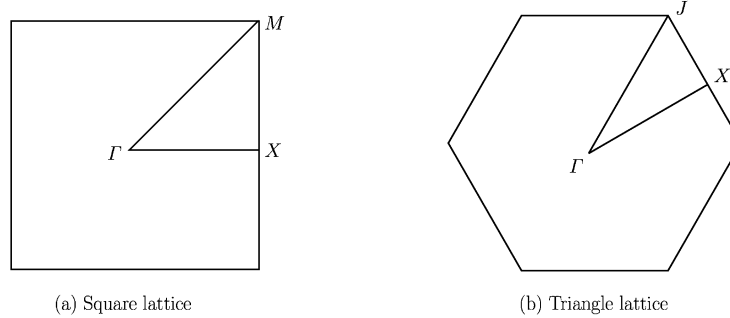
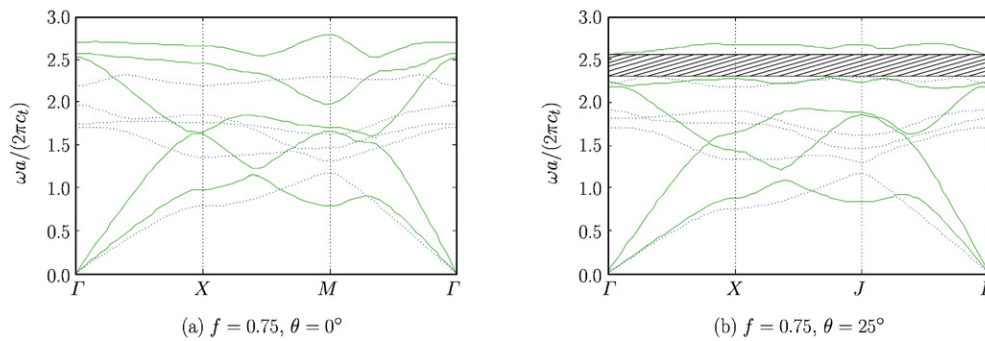


Fig. 2. The first Brillouin zone of 2D lattice.

structure of square lattice is given in Fig.3 for two rotating angles, namely, $\theta = 0^\circ$ and $\theta = 25^\circ$. The solid line denotes the xy mode and the dashed the z mode. The shaded zone denotes the complete band gap. The horizontal axis denotes the wave number. The vertical axis denotes the normalized frequency $\omega a / \pi c_t$, where a is the lattice constants and c_t is the phase velocity of shear wave in the host material. It is noted that no complete band gap exists when $\theta = 0^\circ$ in considered frequency domain. However, a complete band gap appears when $\theta = 25^\circ$. In Fig.4, the wave band structure of triangle lattice array is given for two rotating angles, $\theta = 0^\circ$ and $\theta = 15^\circ$. It is noted that two complete band gaps exist when

Fig. 3. The wave band structure in the case of square lattice array. The xy mode and z mode are indicated by solid and dashed lines, respectively.

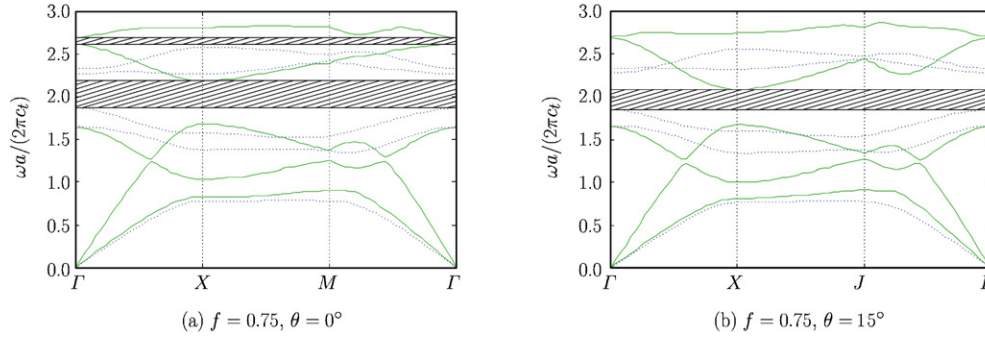


Fig. 4. The wave band structure in the case of triangle lattice array. The xy mode and z mode are indicated by solid and dashed lines, respectively.

$\theta = 0^\circ$ in considered frequency domain. However, a complete band gap only exists when $\theta = 15^\circ$. These observations show that rotating fillers affects evidently the band gap structure. The explanation is that the change of elastic constants leads to the change of wave bands. Similar phenomena were observed in literature [11]. However, it is reported that the z mode is not influenced by rotating fillers. In the present orthotropic case, it is found that z mode is also influenced by rotating fillers. In order to show the influence of rotating fillers on xy mode and z mode clearly, the wave band structure of xy mode and z mode at different rotating angles are given in Fig.5. It is found that both xy mode and z mode are influenced by rotating fillers, although the influence on xy mode is obviously larger than that on z mode. The wave band structure at different rotating angles in the case of triangle lattice array is given in Fig.6 and similar observation is found. These observations can be interpreted as following: The z mode equation in present case of orthotropic filler is related with elastic constants c_{44} , c_{55} , c_{45} and c_{54} . And these elastic constants are dependent upon rotating angle. However, in the case of cubic crystal fillers, z mode equation is only related to elastic constants c_{44} and c_{55} . But c_{44} and c_{55} are independent of the

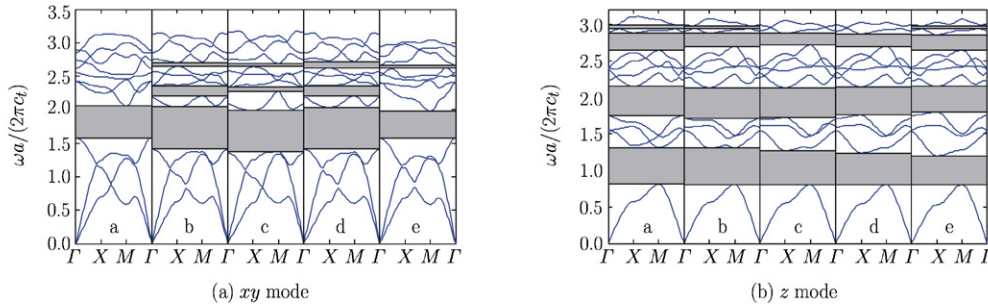


Fig. 5. Band gaps at different rotation angles in the case of square lattice array. There are 10 wave bands and 5 rotation angles are considered. 5 rotation angles, namely, 0° , 30° , 45° , 60° and 90° , are indicated by a, b, c, d and e.

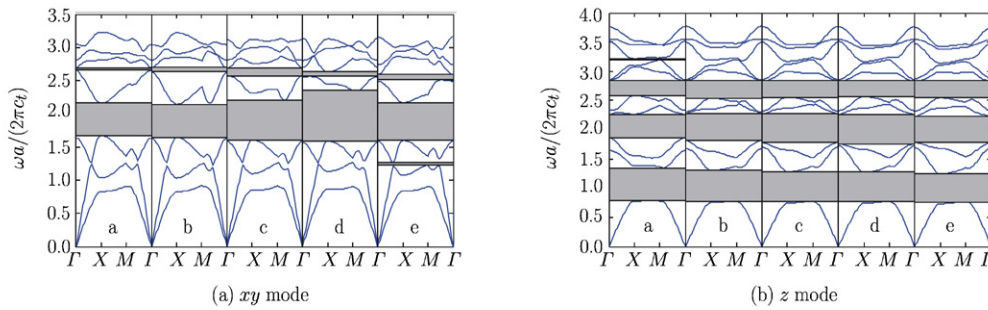


Fig. 6. Band gaps at different rotation angles in the case of triangle lattice array. There are 10 wave band and 5 rotation angles are considered. 5 rotation angles, namely, 0° , 30° , 45° , 60° and 90° , are indicated by a, b, c, d and e.

rotation angle θ in the case of cubic crystal fillers. In addition, the equation of xy mode is related with six elastic constants, namely, c_{11} , c_{12} , c_{16} , c_{22} , c_{26} and c_{66} in contrast that only four elastic constants are related to z mode. These elastic constants are dependent of the rotating angle. Therefore, rotating fillers have larger influence upon xy mode than upon z mode. For anisotropic materials, the anisotropic index, $A = 2c_{44}/(c_{11} - c_{12})$, can be introduced to indicate the anisotropic degree. The numerical computation is carried out for various anisotropic degrees (omitted for abbreviation). It is found that the influence of rotating on wave bands increases gradually with the increase of anisotropic degree of the filler.

V. CONCLUSIONS

The wave band structure of 2D phononic crystal composed of orthotropic fillers embedded in isotropic host is studied. The influence of rotating orthotropic fillers on wave band gaps is investigated numerically in the case of square lattice array and triangle lattice array. It is found that xy modes and z modes are decoupled in the case of orthotropic fillers and both xy mode and z mode are affected by rotating fillers. However, the influence of rotating fillers on xy mode is obviously larger than that on z mode. This phenomenon is observed in both square lattice array and triangle lattice array. This can be ascribed to more elastic constants involved in xy mode equation than in z mode equation. In the case of cubic crystal filler, it is found that rotating fillers does not influence wave band structure of z mode. But, this is only an exception. In general, rotating anisotropic fillers will lead to the change of wave bands of both xy mode and z mode. Therefore, the location and width of complete stop bands can be toned by changing elaborately the rotating angle of anisotropic fillers. In addition, the influence of rotating fillers upon wave bands increases gradually with the increase of anisotropic degree of fillers. In summary, the wave band structure of 2D phononic crystal with anisotropic fillers can be designed elaborately with consideration of the anisotropic feature of fillers except the lattice topology, the material constant contrast and the filling fraction.

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APPENDIX

$$\begin{aligned}
\mathbf{R} &= \left[\rho^{G-G'} \right]_{M \times M} \\
\mathbf{X} \mathbf{M}_1 &= \left[c_{11}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{16}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{61}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{66}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{Y} \mathbf{M}_1 &= \left[c_{16}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{12}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{66}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{62}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{Z} \mathbf{M}_1 &= \left[c_{15}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{14}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{65}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{64}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{X} \mathbf{M}_2 &= \left[c_{61}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{66}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{21}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{26}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{Y} \mathbf{M}_2 &= \left[c_{66}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{62}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{26}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{22}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{Z} \mathbf{M}_2 &= \left[c_{65}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{64}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{25}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{24}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{X} \mathbf{M}_3 &= \left[c_{51}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{56}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{41}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{46}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{Y} \mathbf{M}_3 &= \left[c_{56}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{52}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{46}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{42}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{Z} \mathbf{M}_3 &= \left[c_{55}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_x + c_{54}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_x \right. \\
&\quad \left. + c_{45}^{G-G'} (\mathbf{k} + \mathbf{G}')_x (\mathbf{k} + \mathbf{G})_y + c_{44}^{G-G'} (\mathbf{k} + \mathbf{G}')_y (\mathbf{k} + \mathbf{G})_y \right]_{M \times M} \\
\mathbf{U} \mathbf{X} &= \left\{ u_x^{k+G'_1}, u_x^{k+G'_2}, \dots, u_x^{k+G'_M} \right\}^T \\
\mathbf{U} \mathbf{Y} &= \left\{ u_y^{k+G'_1}, u_y^{k+G'_2}, \dots, u_y^{k+G'_M} \right\}^T \\
\mathbf{U} \mathbf{Z} &= \left\{ u_z^{k+G'_1}, u_z^{k+G'_2}, \dots, u_z^{k+G'_M} \right\}^T
\end{aligned}$$

where $M = 2N + 1$.